THREE HYPOTHESES UNDER WHICH A RELATIVE PROJ IS QUASI-COMPACT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window 3: statement and proof both lie inside the window 6000-8000 granted to this lane by the owner lane lane-s13 on 2026-10-14.

Editorial note. The statement carries the underline-Proj notation of the chapter; it is part of the counted statement span and is preserved verbatim.

Editorial note. The references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Quasi-compactness is local on the base. Schemes, Lemma schemes-lemma-quasi-compact-affine, used at proof lines 7666-7668. Reduces to S affine.
- Construction of the relative Proj by glueing and its affine model. Constructions, Section constructions-section-relative-proj-via-glueing, used at proof lines 7668-7670. Identifies X with Proj(A).
- A_+ is generated by finitely many homogeneous elements. Algebra, Lemma algebra-lemma-S-plus-generated, used at proof lines 7672-7675. Handles case (1).
- Quasi-compactness of Proj of a graded algebra. Constructions, Lemma constructions-lemma-proj-quasi-compact, used at proof lines 7683-7685. Concludes in both cases.

Source reference. Original lemma 07Z4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10134–10134. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01MD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2118–2118. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 01NM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2163–2163. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01K4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2037–2037. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $\mathcal{A}$ be a quasi-coherent graded $\mathcal{O}_S$-algebra. Let $p : X = \underline{Proj}_S(\mathcal{A}) \to S$ be the relative Proj of $\mathcal{A}$. If one of the following holds*

(1) *$\mathcal{A}$ is of finite type as a sheaf of $\mathcal{A}_0$-algebras,*
(2) *$\mathcal{A}$ is generated by $\mathcal{A}_1$ as an $\mathcal{A}_0$-algebra and $\mathcal{A}_1$ is a finite type $\mathcal{A}_0$-module,*
(3) *there exists a finite type quasi-coherent $\mathcal{A}_0$-submodule $\mathcal{F} \subset \mathcal{A}_+$ such that $\mathcal{A}_+/\mathcal{F}\mathcal{A}$ is a locally nilpotent sheaf of ideals of $\mathcal{A}/\mathcal{F}\mathcal{A}$,*

then p is quasi-compact.

Proof. The question is local on the base, see Schemes, Lemma 01K4. Thus we may assume S is affine. Say $S = \mathrm{Spec}(R)$ and $\mathcal{A}$ corresponds to the graded R-algebra A. Then $X = \mathrm{Proj}(A)$, see Constructions, Section 01NM. In case (1) we may after possibly localizing more assume that A is generated by homogeneous elements $f_1, \ldots, f_n \in A_+$ over A_0. Then $A_+ = (f_1, \ldots, f_n)$ by Algebra, Lemma 07Z4. In case (3) we see that $\mathcal{F} = \widetilde{M}$ for some finite type A_0-module $M \subset A_+$. Say $M = \sum A_0 f_i$. Say $f_i = \sum f_{i,j}$ is the decomposition into homogeneous pieces. The condition in (3) signifies that $A_+ \subset \sqrt{(f_{i,j})}$. Thus in both cases we conclude that $\mathrm{Proj}(A)$ is quasi-compact by Constructions, Lemma 01MD. Finally, (2) follows from (1). □